A Desert Plant That Cools Itself by 'Sweating'

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Death Valley in California is one of the hottest places on Earth. In summer, it is often hotter than 49 degrees **Celsius** (120 degrees Fahrenheit). But a **shrub** called the Arizona honeysweet grows well there.

Scientists at Michigan State University found its secret. The plant lets out **water vapor** through tiny holes on its leaves and stems. This is called **transpiration**, and it works a lot like sweating.

The scientists grew more than 1,200 young plants in a lab. Then they slowly made it hotter, up to 60 degrees C (140 degrees F). Some leaves stayed about 13 degrees C (23 degrees F) cooler than the air. When the scientists closed the tiny holes, the plants did not live as long.

"This cooling makes the difference between life and death," said scientist Seung Yon Rhee.

Not every plant survived. Rhee thinks their **DNA** may explain why some lasted longer. The findings have not yet been checked by other scientists.

Word Treasure

shrub -- A small, bushy plant with woody stems.

Celsius -- A scale for measuring temperature.

water vapor -- Water in the form of a gas.

transpiration -- The way plants let out water vapor through their leaves.

DNA -- The instructions inside living things that shape how they grow.

Background you need

Plants take in water through their roots and let some of it out through tiny holes in their leaves.

When water leaves a plant as vapor, it helps cool the plant, a lot like sweat cools your skin.

Death Valley is a very hot, dry desert, so plants there need special tricks to survive.

Break it down

WHO: Scientists at Michigan State University, including Seung Yon Rhee
WHAT: They found the Arizona honeysweet cools its leaves by letting out water vapor
WHERE: Death Valley in California, one of the hottest places on Earth
WHY: The cooling helps the plant survive extreme heat

Why it matters

Just like you sweat to cool down on a hot day, plants have their own ways to beat the heat.

Test what you remember

1. Where does the Arizona honeysweet grow?

(A) Death Valley in California
(B) Under the ocean
(C) A rainforest in Brazil
(D) The North Pole

2. What is transpiration like?

(A) Sweating
(B) Sleeping
(C) Eating
(D) Singing

3. How many young plants did the scientists grow?

(A) Only one
(B) About 12
(C) Exactly 100
(D) More than 1,200

4. How hot did the scientists make the lab?

(A) Up to 60 degrees C (140 degrees F)
(B) Below freezing
(C) Up to 20 degrees C
(D) Up to 100 degrees C

5. What happened when scientists closed the tiny holes?

(A) The plants grew taller
(B) The plants made flowers
(C) Nothing changed
(D) The plants did not live as long

6. What does Seung Yon Rhee think may explain why some plants lasted longer?

(A) Their pots
(B) Their DNA
(C) The color of their leaves
(D) Music in the lab

Answer key (try the quiz first!): 1: A 2: A 3: D 4: A 5: D 6: B

Questions to think about

1. Seung Yon Rhee, scientist: Says the cooling makes the difference between life and death.

MY THOUGHTS (at least 20 words):